+2
+4
+4

10

Institute of Engineering & Technology

BE I Year CS-A & B, CLASS TEST –I, Date 02 12/2022

MER1C3: Elements of Mechanical Engineering

Max. Marks: 20

(60)

Duration: 70 Min.

Note: Attempt all questions.

Q.1 Prove that energy a property of the system 02

Q.2 Derive steady flow energy equation for a boiler (two fluid stream inlet and two fluid stream outlet. 04

Q3 Air flows steadily at the rate of 0.5 kg/s through an air compressor, entering at 7 m/s velocity, 100 kPa pressure and 0.95 m^3/kg volume and leaving at 5 m/s, 700 kPa and 0.19m^3/kg. The internal energy of air leaving is 90 kJ/kg greater than that of air entering. Cooling water in the compressor jackets absorbs heat from the air at the rate of 58 kW. Calculate (a) the rate of shaft work input to the air in kW. (b) ratio of inlet to outlet pipe diameter. 06

Q4 Explain the SAW and SMAW processes with suitable sketches. 04

Q5 Write short notes on (i) spot ERW and continuous spot ERW processes. (ii) Types of electrodes 04

Spot